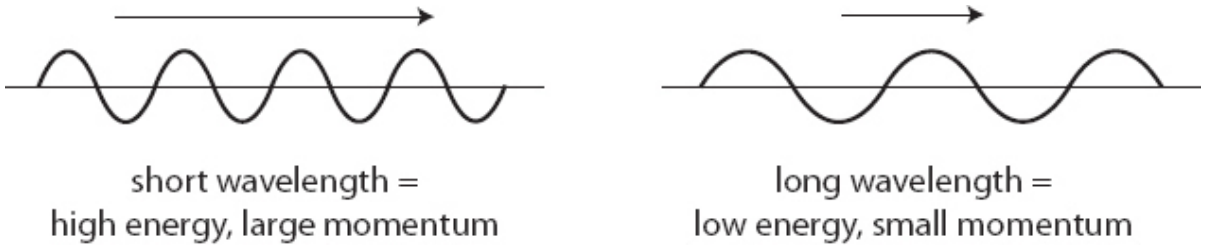
waves. And if that's true, there's no reason we couldn't imagine the same thing going on with electrons—maybe there is something wave-like that carries along the electron particles. That's exactly what de Broglie suggested in his 1924 doctoral thesis, proposing a relationship between the momentum and wavelength of these "matter waves" that was analogous to Planck's formula for light, with larger momenta corresponding to shorter wavelengths.



Like many suggestions at the time, de Broglie's hypothesis may have seemed a little ad hoc, but its implications were far-reaching. In particular, it was natural to ask what the implications of matter waves might be for electrons orbiting around a nucleus. A remarkable answer suggested itself: for the wave to settle down into a stationary configuration, its wavelength had to be an exact multiple of the circumference of a corresponding orbit. Bohr's quantized orbits could be derived rather than simply postulated, simply by associating waves with the electron particles surrounding the nucleus.

Consider a string with its ends held fixed, such as on a guitar or violin. Even though any one point can move up or down as it likes, the overall behavior of the string is constrained by being tied down at either end. As a result, the string only vibrates at certain special wavelengths, or combinations thereof; that's why the strings on musical instruments emit clear notes rather than an indistinct noise. These special vibrations are called the *modes* of the string. The essentially "quantum" nature of the subatomic world, in this picture, comes about not because reality is actually subdivided into distinct chunks but because there are natural vibrational modes for the waves out of which physical systems are made.